

# Confining QCD theory and Mass Gap

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## Abstract

The novel constraint on any solution to QCD has been exactly and uniquely derived in the framework of the the Slavnov-Taylor identities for the gauge particles propagators. It has the two independent solutions. 1). The first one leads to the general QCD, possessing the confinement phase transition. 2).The second one leads to the particular QCD, or, equivalently, the conventional QCD, does not possessing it. The dynamical source of this effect has to be identified with the constant tadpole term, having the dimension of a mass squared, contributing to the full gluon self-energy. Its renormalized version is conventionally called a mass gap. By formulating the proper subtraction scheme, we prove that it cannot be disregarded from the theory and its ground state by any means, but the perturbative renormalizability of QCD will not be affected. The general QCD forbids the free gluons to appear in the physical spectrum at large distances, and, at the time, has an asymptotic freedom behaviour at high energies. Finally, we describe how our advance results are connected to the Jaffe-Witten's theorem.

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## 1. Introduction

The quark model (QM) treats the strongly-interacting hadrons (baryons and mesons) as bound-states of quarks, by emitting and absorbing gluons. The theory, which purpose is to describe the properties of the observed hadrons in terms of the non-observable quarks and gluons from first principles, is Quantum Chromodynamics (QCD) [1, 2, 3, 4, 5, 6, 7, 8, 9]. It is widely accepted as the quantum gauge field theory of strong interactions. Being such a theory, it has to undergo the two phase transitions: the first one at the fundamental (microscopic) quark-gluon level is – the confinement phase transition – in order to explain why all the physical states are colour-singlets. The second one at the (macroscopic) hadronic level is – the PCAC (Partially Conserved Axial Currents) phase transition – in order to explain the soft pion physics [2, 23].

However, this purpose remains a formidable task yet because of the multiple dynamical and topological complexities of low-energy particles (hadrons) physics, originated from the QCD and its ground state (vacuum). This happens because QCD as a gauge theory of the gluon-gluon and quark-gluon strong interactions suffers from a few important conceptual problems. The conventional QCD, which has been investigated the last five decades, cannot explain the colour confinement problem, i.e., why the coloured objects– gluons and quarks– cannot appear at low-energies/large distances, which is an experimental fact.

Secondly, it cannot explain the asymptotic freedom (AF) phenomenon, i.e., a scale breaking at high-energies/short distances [38, 39], which is also an experimental fact. To explain the both experimental facts at low- and high-energies it is necessary to answer a long-standing question how does a mass come out of a massless theory (the mass gap) [2]? even at the fundamental quark gluon level. For example, the linear rising potential between heavy quarks requires the mass scale parameter – the string tension. Also, a scale breaking in AF regime needs a mass squared scale parameter. So that, the goal arises how to solve the above-mentioned problems, and thus to formulate the confining QCD theory, and, at same time, explaining AF phenomenon as well. It may come from the investigation of the true gauge and dynamical structures of the QCD vacuum in more detail.

The properties and symmetries of the QCD Lagrangian, and thus including its Yang–Mills (YM) part, are well-known [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]. The propagation of gluons is one of the main dynamical effects in the QCD ground state. The importance of the corresponding equation of motion is due to the fact that its investigation is supposed to reflect the dynamical and gauge structures of the QCD ground state. The gluon Schwinger–Dyson (SD) equation [2, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] is highly non-linear (NL) because of the self-interaction of massless gluon modes. They are taken into account by the full gluon self-energy, which is the main part of the gluon SD equation itself. We will show that the general properties of the full gluon self-energy point out on some new dynamical and gauge aspects of the true QCD ground

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state. Due to these impacts, the novel insights into a mass dynamical generation in the QCD vacuum are also present within proposed below formalism. It is based on the tensor algebra derivation rigorous rules only, which is widely using in the theoretical and mathematical physics.

## 2. Gluon SD equation

The system of the SD equations of motions and the corresponding Slavnov-Taylor (ST) identities in QCD, in general, contains much more informations on the properties of the theory than its Lagrangian can provide at all [2]. So it might be the place where a mass squared scale parameter could be generated. Among them, the special role belongs to the equations, describing the propagation of the fundamental constituents in the QCD vacuum – gluons and quarks. The above-mentioned gluon SD equation has a rather complicated tensor structure because of its NL character. Let us emphasize that all the known interactions in nature (electroweak, gravitational and strong) are described by the gauge theories. The common mathematical language for them is the tensor algebra, which is only one used throughout this paper. That is why our work will be easily understood by a broad spectrum of researches from the different areas of the theoretical physics. The gluon SD equation analytically looks like

$$D_{\mu\nu}(q) = D_{\mu\nu}^0(q) + D_{\mu\rho}^0(q) i \Pi_{\rho\sigma}(q; D) D_{\sigma\nu}(q), \quad (2.1)$$

where  $D_{\mu\nu}(q)$  and  $D_{\mu\nu}^0(q)$  denote the full gluon propagator and its free counterpart, respectively.  $\Pi_{\rho\sigma}(q; D)$  is the full gluon self-energy which depends on the full gluon propagator due to the non-abelian character of QCD. Here and everywhere below we omit the colour group indices, for simplicity, because of their final factorization, for example  $D_{\mu\nu}^{ab}(q) = D_{\mu\nu}(q) \delta^{ab}$ . Eq. (2.1) in terms of the corresponding skeleton loop diagrams is shown in Fig. 1.

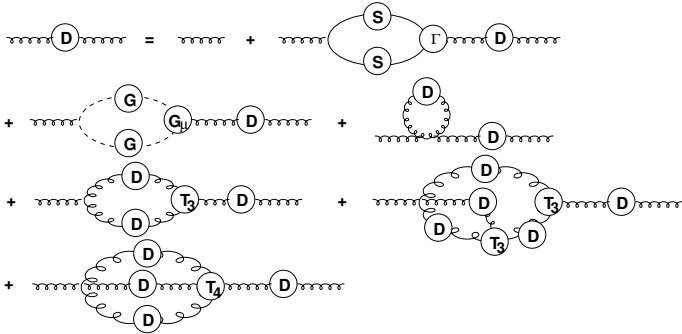


Figure 1: The SDE for the full gluon propagator, as present in [10].

Here stringy lines are for the free gluon propagator, while  $D$  denotes its full counterpart.  $S$  with solid lines denotes the full quark propagator, and  $\Gamma$  denotes the full quark-gluon vertex.  $G$  with dashed lines denotes the full ghost propagator, and  $G_\mu$  is the full ghost-gluon vertex.

Finally,  $T_3$  and  $T_4$  denote the full 3- and 4-gluon vertices, respectively.

The full gluon self-energy is convenient to present as the sum of the three independent terms, namely

$$\Pi_{\rho\sigma}(q; D) = \Pi_{\rho\sigma}^q(q) + \Pi_{\rho\sigma}^g(q; D) + \Pi_{\rho\sigma}^t(D), \quad (2.2)$$

where  $\Pi_{\rho\sigma}^q(q)$  describes the skeleton loop contribution of the quark degrees of freedom as an analogue to the vacuum polarization tensor in Quantum Electrodynamics (QED) [7, 21]. Here and below the superscript or subscript 'q' means quark (not to be mixed up with the gluon momentum variable  $q$ ). The gluon part of the full gluon self-energy by itself is the sum of a few independent terms as follows:

$$\begin{aligned} \Pi_{\rho\sigma}^g(q; D) &= \Pi_{\rho\sigma}^{gh}(q) + \Pi_{\rho\sigma}^{(1)}(q; D^2) \\ &+ \Pi_{\rho\sigma}^{(2)}(q; D^4) + \Pi_{\rho\sigma}^{(2')}(q; D^3), \end{aligned} \quad (2.3)$$

and  $\Pi_{\rho\sigma}^{gh}(q)$  describes the skeleton loop contribution associated with the ghost degrees of freedom.  $\Pi_{\rho\sigma}^{(1)}(q; D^2)$  represents the skeleton loop contribution, containing the 3-gluon vertices only. Finally,  $\Pi_{\rho\sigma}^{(2)}(q; D^4)$  and  $\Pi_{\rho\sigma}^{(2')}(q; D^3)$  describe the skeleton two-loop contributions, which combine the 3- and 4-gluon vertices. All these quantities are given by the corresponding skeleton loop diagrams in Fig. 1, and they are independent from each other. The last four terms explicitly contain the full gluon propagators in the corresponding powers symbolically shown above. The analytical expressions for the corresponding skeleton loop integrals [22], in which the symmetry and combinatorial coefficients and signs have been included, are not important here. We are not going to calculate any of them explicitly, and thus to introduce into them any truncations/approximations/assumptions or choose some special gauge. Any skeleton loop integral in Fig. 1 is the sum of the infinite number of terms. Moreover, the full vertices entering these skeleton loop integrals are themselves determined by an infinite series of the corresponding multi-loop skeleton terms [2, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]. In these NL series the dependence on the coupling constant may not be simple, i.e., in fact, it is not known. Such kind of series are the so-called 'cluster' expansions [23], indeed.

The constant tadpole term  $\Pi_{\rho\sigma}^t(D)$  is defined as,

$$\Pi_{\rho\sigma}^t(D) \sim \int d^4l D_{\alpha\beta}(l) T_{\rho\sigma\alpha\beta}^0 = g_{\rho\sigma} \Delta_t^2(D), \quad (2.4)$$

where  $g_{\rho\sigma} = T_{\rho\sigma}(q) + L_{\rho\sigma}(q) = T_{\rho\sigma}(q) + q_\rho q_\sigma / q^2$ . In what follows the constant  $\Delta_t^2(D)$  will be called as the tadpole term itself, since it deviates from the definition (2.4) by the tensor  $g_{\rho\sigma}$  only. So that, the tadpole term equally contributes to the transverse and longitudinal components of the full gluon self-energy (2.2).

It is instructive for further purpose to present the initial gluon SD eq. (2.1), on account of the expressions (2.2) and (2.4) as follows:

$$D_{\mu\nu}(q) = D_{\mu\nu}^0(q) + D_{\mu\rho}^T(q) + D_{\mu\rho}^L(q) \quad (2.5)$$

with

$$\begin{aligned} D_{\mu\nu}^T(q) &= D_{\mu\rho}^0(q) i [\Pi_{\rho\sigma}^q(q) + \Pi_{\rho\sigma}^g(q; D)] D_{\sigma\nu}(q), \\ D_{\mu\nu}^L(q) &= D_{\mu\rho}^0(q) i g_{\rho\sigma} \Delta_t^2(D) D_{\sigma\nu}(q). \end{aligned} \quad (2.6)$$

All the terms which contribute to the full gluon self-energy eq. (2.2), and hence eq. (2.5), are tensors, having the dimensions of a mass squared. All these skeleton loop integrals are therefore quadratically divergent (QD), i.e., ultraviolet (UV) divergent in the perturbation theory (PT) regime, and so they are assumed to be regularized from above. Contrary to QED, QCD being a non-abelian gauge theory can suffer from the severe infrared (IR) singularities in the  $q^2 \rightarrow 0$  limit (i.e., more singular than the PT one, which is proportional to  $\sim 1/q^2$ ). Evidently, they will be generated due to the self-interaction of massless gluon modes. Thus, all the skeleton loop integrals, contributing to the full gluon self-energy, are assumed to be regularized from below as well. This means that all the expressions are regularized (i.e., we render infinite skeleton loop integrals to be finite). How precisely the regulating parameters have to be introduced is not important here, since we are not going to calculate any of these skeleton loop integrals explicitly, as emphasized above. They will be assumed but not shown in all regularized expressions, for simplicity.

The constant tadpole term  $\Delta_t^2(D)$ , defined in eq. (2.4), is nothing else but the QD, i.e., UV divergent skeleton loop integral, which is already regularized from below and above, as well as all the other such kind of constants which may appear in the theory by any possible ways (the subtraction schemes, the iteration solutions, etc., see below). Being a mass squared regularized quantity, it is explicitly present in the QCD ground state, see eq. (2.5). Then a natural question arises why is it present in the vacuum of QCD at all if it makes the theory to look non-renormalizable from the very beginning? Let us remind that the theory possessing the QD quantities is not the PT renormalizable one since the logarithmic-type divergences only can be accounted for the PT. The standard solution was to remove this and all other QD constants in any possible gauge-invariant way from the theory in order to make it renormalizable in the PT sense. **However, here we are going to ask the question is it possible to retain the tadpole term in the QCD vacuum, but without affecting the PT renormalizability of the theory?** This is important to ask because, unlike to other QD constants, the tadpole term is only one which dynamically generates a mass squared scale parameter, dominated by the point-like four-gluon vertex only, and thus does not depending on the external gluon momentum at all [6]. Also, we will clarify its role in the dynamical and gauge structures of the QCD ground state in more detail.

Concluding, let us remind that the whole gluon momentum range is  $q^2 \in [0, \infty)$ . In what follows we will work in Euclidean metric  $q^2 = q_0^2 + \mathbf{q}^2$  since it implies  $q_i \rightarrow 0$

when  $q^2 \rightarrow 0$  and *vice versa*. This makes it possible to avoid the unphysical IR singularities on the light cone.

### 3. Exact constraint on any solution to QCD

It is well known that all the ST identities which appear in QCD are important for the renormalizability of the theory, and thus they "are exact constraints on any solution to QCD" [2]. Among them the ST identities [2, 24, 35] for the gauge particle propagators play a dominant role, since they determine the gauge structure of the QCD vacuum.

The ST identity for the full gluon propagator looks like

$$q_\mu q_\nu D_{\mu\nu}(q) = i\xi, \quad (3.7)$$

where  $\xi$  is the gauge-fixing parameter. It implies that the general tensor decomposition of the full gluon propagator in the covariant gauge is as follows:

$$D_{\mu\nu}(q) = i [T_{\mu\nu}(q) d(q^2) + \xi L_{\mu\nu}(q)] \frac{1}{q^2}, \quad (3.8)$$

where the invariant function  $d(q^2)$  is the corresponding Lorentz structure of the full gluon propagator (or the gluon invariant function). Throughout this paper we use the standard definition of  $T_{\mu\nu}(q) = \delta_{\mu\nu} - q_\mu q_\nu / q^2 = \delta_{\mu\nu} - L_{\mu\nu}(q)$  in Euclidean metric. Any invariant functions associated with the projection operators  $T_{\mu\nu}(q)$  and  $\delta_{\mu\nu}$  are the same, and thus  $D_{\mu\nu}(q)$  is defined up to its longitudinal part  $L_{\mu\nu}(q)$ . This should be also valid for its equation of motion. By formally setting  $d(q^2) = 1$  in eq. (3.8), then one obtains the free gluon propagator

$$D_{\mu\nu}^0(q) = i [T_{\mu\nu}(q) + \xi_0 L_{\mu\nu}(q)] \frac{1}{q^2}, \quad (3.9)$$

where  $\xi_0$  is the corresponding gauge-fixing parameter. The general ST identity (3.7) will look like

$$q_\mu q_\nu D_{\mu\nu}^0(q) = i\xi_0. \quad (3.10)$$

It is important to note in advance that from the very beginning the gauge-fixing parameter for the full gluon propagator  $\xi$  is not to be identified with its free counterpart  $\xi_0$ . Since the gauge freedom in QCD is determined by  $\xi_0$ , there should exist a relation between them.

Let us now perform some exact algebraic derivations which are necessary for the further purposes. Contracting the full gluon SD eq. (2.1) with  $q_\mu$  and  $q_\nu$ , on account of the relations (3.7)-(3.10), one gets

$$q_\rho q_\sigma \Pi_{\rho\sigma}(q; D) = \frac{(\xi_0 - \xi)}{\xi \xi_0} (q^2)^2, \quad (3.11)$$

where the full gluon self-energy, due to the eqs. (2.2) and (2.4), can be present as follows:

$$\Pi_{\rho\sigma}(q; D) = \Pi_{\rho\sigma}^q(q) + \Pi_{\rho\sigma}^g(q; D) + \delta_{\rho\sigma} \Delta_t^2(D), \quad (3.12)$$

and contracting it with  $q_\rho$  and  $q_\sigma$ , and because of the previous relation (3.11), one obtains

$$q_\rho q_\sigma [\Pi_{\rho\sigma}^q(q) + \Pi_{\rho\sigma}^g(q; D)] = \frac{(\xi_0 - \xi)}{\xi \xi_0} (q^2)^2 - q^2 \Delta_t^2(D). \quad (3.13)$$

Obviously, this transverse relation can be obtained by the corresponding contraction of eq. (2.5). It explicitly depends on the tadpole term  $\Delta_t^2(D)$ . If it is formally put zero, i.e., by hand, then the both transverse relations (3.11) and (3.13) coincide with each other. This underlines its important role in the QCD ground state, which one precisely is to be clarified in our paper. The transverse relation (3.11) and the transverse relation obtained from eq. (3.12) are independent from each other, since the first one has been derived by using the gluon SD eq. (2.1), while the second one by using the initial definition (2.2). From all these transverse relations one concludes that by themselves they cannot remove the QD constants which may appear in the theory (in order to make it in the PT sense renormalizable), and to fix the relation between  $\xi$  and  $\xi_0$ .

Let us now remind that in QCD the quark skeleton loop contribution into the full gluon self-energy can be always made transverse because of the electric charge conservation which flows around the closed loop (in complete analogy with QED, see [10] and derivations below). Therefore,  $q_\rho q_\sigma \Pi_{\rho\sigma}^q = 0$  and this relation holds independently from the gluon contribution, which has been defined in eq. (2.3). At the same time, it is well-known that in QCD just the Faddeev–Popov (FP) ghost contribution [42]  $\Pi_{\rho\sigma}^{gh}(q)$  makes the transverse relation for this term satisfied (independently from the tadpole term (2.4), which itself has the transverse projection operator structure). It is instructive to emphasize that each term in the sum (2.3) cannot be made transverse, only the whole sum. How explicitly this works can be found in any text book on QCD. For the most transparent calculations up to one loop contribution to the transverse gluon self-energy (2.3) see for example [3, 4, 5, 6]. Of course, such a transversality will be achieved in the higher loop iteration terms as well, i.e., it is a general fact, as it is requested by the satisfied transverse relation for the (2.3) term. It is a sum of the skeleton loop contributions, as underlined from the very beginning in this work. This means that  $q_\rho q_\sigma \Pi_{\rho\sigma}^g(q; D) = 0$  and this relation holds independently from the quark contribution.

So that, in general, in QCD one has

$$q_\rho q_\sigma \Pi_{\rho\sigma}^q(q) = q_\rho q_\sigma \Pi_{\rho\sigma}^g(q; D) = 0, \quad (3.14)$$

and then from the relation (3.13), one immediately arrives at the exact condition, namely

$$\frac{(\xi_0 - \xi)}{\xi \xi_0} q^2 = \Delta_t^2(D), \quad (3.15)$$

which can be treated as the exact constraint on any solution to QCD, since coming out from the corresponding ST identities, and deriving before going to the formulation of any proper subtraction scheme. This underlines its intrinsic, independent and important status in the theory.

### 3.1. General solution

The general solution of the exact constraint (3.15) determines the function  $\xi = f(q^2; \xi_0)$  as follows:

$$\xi = f(q^2; \xi_0) = \frac{\xi_0 q^2}{q^2 + \xi_0 \Delta_t^2(D)}. \quad (3.16)$$

This relation has been exactly (i.e., without making any kind of the simplifications) and uniquely defined in a such new manner for the first time. This expression demonstrates the NL dependence  $\xi = f(q^2; \xi_0)$  on  $\xi_0$ . The linear relation  $\xi \rightarrow \xi_0$  will be recovered in the PT  $q^2 \rightarrow \infty$  limit, when the ratio  $(\Delta_t^2(D)/q^2)$  is to be suppressed in this regime at finite  $\xi_0$ . We have established the general boundary condition in QCD for the behaviour of this relation in the PT limit for the regularized full gluon propagator. Since the exact solution (3.16) has been obtained within the ST identities general formalism, we will call it as the general solution to QCD and its vacuum, i.e., at the fundamental quark-gluon level (see section 5 for detailed description).

### 3.2. Particular solution

The exact constraint (3.15) has also a particular solution, namely

$$\Delta_t^2(D) = 0 \rightarrow \xi = \xi_0, \text{ and vice versa}, \quad (3.17)$$

but these equalities should be put by hand in this case. They are the prescriptions rather than the exact mathematical results. In what follows we will call it as conventional or, equivalently, the PT QCD as well (though the coupling constant remains always strong apart from the asymptotic free (AF) regime [2, 38, 39]). For its description in some more details see section 6.

## 4. Proper subtraction scheme

QCD being much more complicated quantum field gauge theory than QED, has the two independent satisfied transverse relations (3.14), and so requires much more careful investigation in QCD. They have to be included in a self-consistent way into the proper subtraction scheme. Finally this will lead to the removal of the corresponding QD constants from the theory. How they fixed the relations (3.16) and (3.17) in the most general and unique ways has been just described above.

The first step in the renormalization program of any gauge theory is the removal of the quadratic UV divergences in order to make the corresponding theory renormalizable in the PT sense. It can be achieved by introducing the proper subtraction scheme in order to separate them from the PT logarithmic divergences. The preliminary step in the regularization program has been already done by introducing the corresponding regulating parameters, mentioned above. Within our approach nothing will

depend on how exactly the regulating parameters have been introduced. Even the hard cut-off procedures can be used, since we are mainly interested in the regulating of leading order UV divergences [2]. They have to disappear from the theory after the PT and the non-perturbative (NP) renormalization programs will be performed. Let us now introduce the subtractions for the quark and gluon contributions to the full gluon self-energy as follows:

$$\Pi_{\rho\sigma}^{s(q)}(q) = \Pi_{\rho\sigma}^q(q) - \Pi_{\rho\sigma}^q(0) = \Pi_{\rho\sigma}^q(q) - \delta_{\rho\sigma}\Delta_q^2, \quad (4.18)$$

$$\begin{aligned} \Pi_{\rho\sigma}^{s(g)}(q; D) &= \Pi_{\rho\sigma}^g(q; D) - \Pi_{\rho\sigma}^g(0; D) \\ &= \Pi_{\rho\sigma}^g(q; D) - \delta_{\rho\sigma}\Delta_g^2(D), \end{aligned} \quad (4.19)$$

and thus  $\Pi_{\rho\sigma}^{s(q)}(0) = \Pi_{\rho\sigma}^{s(g)}(0; D) = 0$ , by definitions. In these relations  $\Delta_q^2$  is the skeleton loop integral for the quark degrees of freedom at  $q = 0$ , and  $\Delta_g^2(D)$  is the sum of the corresponding skeleton loop integrals at  $q = 0$  contributing to eq. (2.3), namely

$$\Delta_g^2(D) = \Delta_{gh}^2 + \Delta_1^2(D^2) + \Delta_2^2(D^4) + \Delta_{2'}^2(D^3). \quad (4.20)$$

Let us remind that all the QD constants, shown up in eqs. (4.18)-(4.19), as well as the tadpole term  $\Delta_t^2(D)$  itself, are independent from each other and are defined by the skeleton loop integrals, which have been already regularized from above and below. The subtraction at zero is to be understood in a such way that we subtract at the external gluon momentum  $q^2 = -\mu^2$  [2] with  $\mu^2 \rightarrow 0$  final limit. However, due to the self-interaction of the massless gluon modes, some of these constants (apart from the tadpole term) may depend on the internal zero gluon momentum (if it is connected to any closed loop, for example such as which can appear in the iteration solution of the gluon SD equation [6]). Then it has to be replaced by the subtraction at  $q_i^2 = -M_i^2$  with  $M_i^2 \rightarrow 0$  final limit as well, and subindex  $i$  determines the number of such internal gluons. For the any single full gluon propagator the subindex  $i$  can be treated as the number of the necessary subtractions made in it in accordance with the rules of the theory of distributions (generalized functions) [26].

Also it is necessary to emphasize that the subtractions (4.18)-(4.19) have not been introduced by hand. They are equivalent to add zero to the corresponding identity. For example  $\Pi_{\rho\sigma}^q(q) = \Pi_{\rho\sigma}^q(q) - \Pi_{\rho\sigma}^q(0) + \Pi_{\rho\sigma}^q(0)$ , so that the initial quark contribution to the gluon self-energy is not changed. It makes it possible to present the initial quark contribution  $\Pi_{\rho\sigma}^q(q) = \Pi_{\rho\sigma}^{s(q)}(q) + \Pi_{\rho\sigma}^q(0)$  as a sum of the two terms, one of which shows up all the corresponding QD but regularized constants. The first term is a regular function of the external gluon momentum  $q$ , by definition, and it may be only logarithmically divergent at large  $q$ . In other words, we separate the NP QD constants from the PT logarithmically divergent terms in a gauge-invariant way, that is why our subtraction scheme is called as a

proper one. The same is true for the initial gluon contribution as well. Such an exact separation will be very useful for the renormalization programs of any kind to be performed for the single full gluon propagator, and we consider this as an advantage of the formalism developed here. For the interacting full gluon propagators these independent terms will interact with each other, but such a separation will simplify the corresponding renormalization programs, anyway.

The independent tensor decompositions of the quark and gluon degrees of freedom, which appear in the subtraction relations (4.18)-(4.19), are

$$\begin{aligned} \Pi_{\rho\sigma}^{s(q)}(q) &= T_{\rho\sigma}(q)q^2\Pi_t^{s(q)}(q^2) - q_\rho q_\sigma \Pi_l^{s(q)}(q^2), \\ \Pi_{\rho\sigma}^q(q) &= T_{\rho\sigma}(q)q^2\Pi_t^q(q^2) - q_\rho q_\sigma \Pi_l^q(q^2), \end{aligned} \quad (4.21)$$

and

$$\begin{aligned} \Pi_{\rho\sigma}^{s(g)}(q) &= T_{\rho\sigma}(q)q^2\Pi_t^{s(g)}(q^2) - q_\rho q_\sigma \Pi_l^{s(g)}(q^2), \\ \Pi_{\rho\sigma}^g(q) &= T_{\rho\sigma}(q)q^2\Pi_t^g(q^2) - q_\rho q_\sigma \Pi_l^g(q^2), \end{aligned} \quad (4.22)$$

respectively. In all the quantities above and below the dependence on  $D$  is omitted, for simplicity, and will be restored when necessary. Here and everywhere below all the invariant functions are dimensionless ones of their argument  $q^2$ : otherwise they remain arbitrary. However, all the invariant functions  $\Pi_t^{s(q)}(q^2)$ ,  $\Pi_l^{s(q)}(q^2)$  as well as  $\Pi_t^{s(g)}(q^2)$ ,  $\Pi_l^{s(g)}(q^2)$  cannot have the power-type singularities (or, equivalently, the pole-type or the massless ones) at small  $q^2$ , by definitions, as it follows from the initial subtractions (4.18)-(4.19).

Substituting all these decompositions (4.21)-(4.22) into the subtractions (4.18)-(4.19), and doing some tensor algebra derivations, one finally obtains

$$\begin{aligned} \Pi_t^{s(q)}(q^2) &= \Pi_t^q(q^2) - \frac{\Delta_q^2}{q^2}, \\ \Pi_l^{s(q)}(q^2) &= \Pi_l^q(q^2) + \frac{\Delta_q^2}{q^2}, \end{aligned} \quad (4.23)$$

and

$$\begin{aligned} \Pi_t^{s(g)}(q^2) &= \Pi_t^g(q^2) - \frac{\Delta_g^2(D)}{q^2}, \\ \Pi_l^{s(g)}(q^2) &= \Pi_l^g(q^2) + \frac{\Delta_g^2(D)}{q^2}. \end{aligned} \quad (4.24)$$

Using these relations it is easy to show that for the regularized (i.e., finite) quark and gluon contributions to the initial gluon self-energy their values at zero always are  $\Pi_{\rho\sigma}^q(0; D) = \delta_{\rho\sigma}\Delta_q^2$  and  $\Pi_{\rho\sigma}^g(0; D) = \delta_{\rho\sigma}\Delta_g^2(D)$ , indeed, i.e., they are general ones, and have been already used in the subtraction relations (4.18)-(4.19) above, as it has to

be. Also these relations will be true if expressed in terms of the above-mentioned subtraction point  $q^2 = -\mu^2$ .

An interesting observation follows from the relations (4.23) and (4.24), namely  $\Pi_t^q(q^2) + \Pi_l^q(q^2) = \Pi_t^{s(q)}(q^2) + \Pi_l^{s(q)}(q^2)$  and  $\Pi_t^g(q^2) + \Pi_l^g(q^2) = \Pi_t^{s(g)}(q^2) + \Pi_l^{s(g)}(q^2)$ . So that the corresponding sums do not depend on the corresponding QD but regularized constants, which once more underlines the general character of our subtraction scheme. Also, the sum of the subtracted invariant functions is never zero, though the subtracted contributions to the full gluon self-energy are zero at  $q^2 = 0$ , by definitions, see relations (4.18)-(4.19).

As we already know, the quark and gluon contributions to the full gluon self-energy are transverse  $q_\rho q_\sigma \Pi_{\rho\sigma}^q(q) = q_\rho q_\sigma \Pi_{\rho\sigma}^g(q; D) = 0$ , see the relations (3.14). Then from the second of the relations (4.21) and (4.22) it follows that  $\Pi_l^q(q^2) = \Pi_l^g(q^2) = 0$  as well. So that the second relations in (4.23) and (4.24) will be reduced to

$$\Pi_l^{s(q)}(q^2) = \frac{\Delta_q^2}{q^2}, \quad \Pi_l^{s(g)}(q^2; D) = \frac{\Delta_g^2(D)}{q^2}, \quad (4.25)$$

where we restored the dependence on  $D$  in the second relation. However, these relations are impossible, since the corresponding invariant functions cannot have the pole-type singularities, by definitions, as explained above. One has to put these constants to zero on this general mathematical basis, i.e.,  $\Delta_q^2 = \Delta_g^2(D) = 0$ , and thus  $\Pi_l^{s(q)}(q^2) = \Pi_l^{s(g)}(q^2; D) = 0$  as well. All our results for the QD constants can be now summarized as follows:

$$\Delta_q^2 = \Delta_g^2(D) = 0, \quad \Delta_t^2(D) \neq 0, \quad (4.26)$$

and hence  $\Delta_{gh}^2 = \Delta_1^2(D^2) = \Delta_2^2(D^4) = \Delta_{2'}^2(D^3) = 0$ , as well because of the relation (4.20). The system of the relations (4.26) is not only a general but it is a unique one as well. They are not prescriptions, since based on a rigorous tensor algebra derivation rules, i.e., they are exact mathematical results. The characteristic feature of this system is that the tadpole term  $\Delta_t^2(D)$  remains intact. The mathematical reason is that for the constant tadpole term such kind of the non-trivial relations (4.18)-(4.19) does not exist (constant minus the same constant is always zero). The essential dynamical source of its survival in the theory is that only the tadpole term can generate a mass squared scale parameter [6]. All the other quark and gluon terms, contributing to the full gluon self-energy, cannot do this, and thus none of their subtracted counterparts with their quark and gluon constants.

All these constants, defined by the corresponding skeleton loop integrals, are QD at the upper limit. They have to be removed/disregarded from the theory on the general mathematical basis, i.e., put zero, as described above, apart from the tadpole term (2.4). In comparison with it, all the other QD quark and gluon constants will be called the tadpole-like/type terms. The general question

arises now, namely how to understand these exact equalities to zero in the relations (4.26)? These equalities mean that any tadpole-like term which may appear in the theory by any possible way has to be discarded/disregarded in the theory, i.e., put zero, independently from any other tadpole-like terms. In this way such sums will be always zero, indeed. Such constants may appear even not as the result of the subtractions, but, for example as the result of the NL iteration procedure for the full gluon propagator [6] (see appendix A as well). In this case the tadpole-like terms may be even multiplied by some regularized functions. Whatever their origins would have been in the full gluon propagator, all of them belong to the infinite manifold of the relations (4.26), and thus should be always removed from the theory.

Collecting our results obtained above for the quark and gluon contributions to the gluon self-energy, one obtains

$$\begin{aligned} \Pi_{\rho\sigma}^q(q) &= T_{\rho\sigma}(q) q^2 \Pi_t^{s(q)}(q^2), \\ \Pi_{\rho\sigma}^g(q; D) &= T_{\rho\sigma}(q) q^2 \Pi_t^{s(g)}(q^2; D), \end{aligned} \quad (4.27)$$

since they become transverse and coincide with their subtracted counterparts. Reminding also that the invariant functions present in the previous relations are regular functions at small  $q^2$ , i.e., they have no pole-type singularities, and may be only logarithmically divergent at large  $q^2$ . This becomes possible only due to the satisfied transverse relations (3.14) for the quark and gluon degrees of freedom. Just this decreases the quadratic UV divergences of the corresponding skeleton loop integrals to a logarithmic ones, as it has been described in detail in this section.

## 5. General QCD

Substituting the sum of the terms (4.27) into the eq. (3.12), on account of the relation  $\delta_{\rho\sigma} = T_{\rho\sigma}(q) + L_{\rho\sigma}(q)$  and further into the gluon SD eq. (2.5), one finally gets

$$D_{\mu\nu}(q) = D_{\mu\nu}^0(q) + D_{\mu\nu}^T(q; D) + D_{\mu\nu}^L(q; D), \quad (5.28)$$

where

$$D_{\mu\nu}^T(q; D) = D_{\mu\rho}^0(q) i T_{\rho\sigma}(q) [q^2 \Pi(q^2; D) + \Delta_t^2(D)] D_{\sigma\nu}(q), \quad (5.29)$$

and

$$D_{\mu\nu}^L(q; D) = D_{\mu\rho}^0(q) i L_{\rho\sigma}(q) \Delta_t^2(D) D_{\sigma\nu}(q), \quad (5.30)$$

respectively.  $\Pi(q^2; D) = \Pi_t^{s(q)}(q^2) + \Pi_t^{s(g)}(q^2; D)$ , and this invariant function is regular at zero and may have only the logarithmic divergences in the PT  $q^2 \rightarrow \infty$  limit. The gluon SD eq. (5.28) is now equivalent to the initial gluon SD eq. (2.5), only being re-written in the different form.

Combining the general gluon SD eq. (5.28) with the decompositions (1.2) and (1.3), and on account of the relations (5.29) and (5.30), one obtains

$$d(q^2) = \frac{1}{1 + \Pi(q^2; D) + (\Delta_t^2(D)/q^2)}. \quad (5.31)$$

This relation is the NL transcendental equation for the different invariant functions  $d(q^2)$ ,  $\Pi(q^2; D)$  and the constant  $\Delta_t^2(D)$ , i.e.,  $d = f(D(d))$ . Nevertheless, from this expression is clearly seen that in the PT  $q^2 \rightarrow \infty$  regime, the contribution  $(\Delta_t^2(D)/q^2)$  can be neglected, but the invariant function  $\Pi(q^2; D)$  may still depend on this ratio under the PT logarithms. In the NP region of finite and small gluon momenta this term is dominant, and the dependence of  $d(q^2)$  on  $\Delta_t^2(D)$  may be much more complicated due to the transcendental character of eq (5.31).

Contracting the full gluon SD eq. (5.28), on account of the relation (5.30), with  $q_\mu$  and  $q_\nu$ , and substituting its result into the general ST identity (3.7), one arrives at

$$q_\mu q_\nu D_{\mu\nu}(q) = i\xi_0 \left( 1 - \xi \frac{\Delta_t^2(D)}{q^2} \right) = i\xi, \quad (5.32)$$

which solution is

$$\xi \equiv \xi(q^2; \xi_0) = \frac{\xi_0 q^2}{q^2 + \xi_0 \Delta_t^2(D)}, \quad (5.33)$$

i.e., in this case the gauge-fixing parameter becomes the function  $\xi \equiv \xi(q^2; \xi_0)$  and thus it is not a constant equal to  $\xi_0$ . Let us point out that the expression (5.33) coincides with the general solution (3.16), which is as it should be. So that, its asymptotic properties have been already discussed in the text after eq. (3.16). The function (5.33) is the known function of its arguments, while the relation (5.31) is the NL transcendental one. Behind the general inequality  $\xi \neq \xi_0$  is the regularized constant  $\Delta_t^2(D)$  as its dynamical source.

Substituting equations (5.31) and (5.33) into the general decomposition (3.8) for the full gluon propagator, one finally obtains

$$\begin{aligned} D_{\mu\nu}(q) &= iT_{\mu\nu}(q) \frac{1}{q^2 + q^2 \Pi(q^2; D) + \Delta_t^2(D)} \\ &+ iL_{\mu\nu}(q) \frac{\lambda^{-1}}{q^2 + \lambda^{-1} \Delta_t^2(D)}, \end{aligned} \quad (5.34)$$

where we have introduced the useful notation  $\xi_0 = \lambda^{-1}$  [4]. The corresponding ST identity now becomes

$$q_\mu q_\nu D_{\mu\nu}(q) = i\xi(q^2; \lambda^{-1}) = i \frac{\lambda^{-1} q^2}{q^2 + \lambda^{-1} \Delta_t^2(D)}. \quad (5.35)$$

The ST identity (5.35) depends on the constant  $\Delta_t^2(D)$ , and when it is zero, one recovers the gauge-fixing parameter for the free gluon propagator. In this ST identity the gauge-fixing parameter  $\lambda^{-1}$  for the free gluon propagator is convenient to vary continuously from zero to infinity. The functional dependence of  $\xi$  (5.33) is fixed up to an arbitrary gauge-fixing parameter  $\xi_0 = \lambda^{-1}$ . Unless we fix it, and thus  $\xi$  itself, we will call such situation as the generalized gauge dependence (GGD), see equations (3.7)-(3.8), (5.32)-(5.35). Choosing  $\xi_0 = \lambda^{-1}$  explicitly, we will call such situation as the explicit gauge dependence (EGD).

For example,  $\xi_0 = \lambda^{-1} = 0$  is called the unitary (Landau) gauge,  $\xi_0 = \lambda^{-1} = 1$  is called the t' Hooft-Feynman gauge, etc. [30, 31, 32, 33]. The formal  $\xi_0 = \lambda^{-1} = \infty$  limit is called as the canonical gauge in [32]. This distinction seems a mere convention, but, nevertheless, it is useful one in QCD because of the presence of a mass scale parameter in its ground state. The generalized gauge requires that there is no other functional expression for  $\xi$ , apart from given by the relation (5.33) at finite  $\xi_0 = \lambda^{-1}$ , in the full gluon propagator (5.34) and the ST identity (5.35) for the regularized gluon fields. For the renormalized version of the ST identity (5.35) see section 6 below.

The system of the regularized equations (5.34)-(5.35), explicitly depending on the tadpole term, are present in the form suitable for the NP renormalization program to be performed. The tadpole term enters the full gluon self-energy linearly, see the expression (3.12). However, in the full gluon propagator (5.34) it appears in the NL way, because its contribution has been summed up with the help of the gluon SD eq. (5.28). The term  $(\Delta_t^2(D)/q^2)$  in the expressions (5.34)-(5.35) is to be suppressed in the PT  $q^2 \rightarrow \infty$  limit at the finite  $\xi_0 = \lambda^{-1}$  (the above-mentioned canonical gauge will be investigated in detail in the forthcoming paper). Then the full gluon propagator will behave like the free gluon propagator  $\sim 1/q^2$  in this limit. This is the one of the necessary constraints that a theory is perturbative renormalizable. Other ones such as the corresponding behaviour of the spinor Green's function, a unitarity of  $S$ -matrix and analyticity (causality) [2], evidently are beyond the scope of the present work. Here one can conclude that the true dynamical and gauge structures of the QCD/YM ground state are much more complicated than it follows from its Lagrangian's formulation. However, its PT renormalizability is not affected within the QCD full gluon propagator in the generalized gauge (5.34), i.e., it has the PT renormalizable behaviour at large  $q^2$ .

Concluding, it is worth noting that in terms of the transverse relations and the corresponding QD regularized constants as well as the corresponding relations between  $\xi$  and  $\xi_0$ , the general solution can be shown explicitly as the following system of the relations, namely

$$\begin{aligned} q_\rho q_\sigma \Pi_{\rho\sigma}^q(q) &= q_\rho q_\sigma \Pi_{\rho\sigma}^g(q; D) = 0, \\ q_\rho q_\sigma \Pi_{\rho\sigma}(q; D) &= q^2 \Delta_t^2(D), \end{aligned} \quad (5.36)$$

and

$$\Delta_q^2 = \Delta_g^2(D) = 0, \quad \Delta_t^2(D) \neq 0, \quad \xi \neq \xi_0 = \lambda^{-1}. \quad (5.37)$$

It is necessary to underline that none of the prescriptions (i.e., putting some of them zero by hand) have been introduced and none of the truncations, approximations and assumptions have been done, as well as no gauge fixing by hand, in the obtaining of these relations. In other words, all of them are exact mathematical results in the general solution to QCD.

## 6. Renormalized ST identity and mass gap

Since the novel constraint derived in section 3 has been mainly based on the ST identities formalism, it is possible to perform the NP renormalization program for the ST identity (5.35) itself. It is instructive to present (5.35) in terms of  $\xi$  and  $\xi_0$  again, then we have

$$q_\mu q_\nu D_{\mu\nu}(q) = i\xi = i \frac{\xi_0 q^2}{q^2 + \xi_0 \Delta_t^2(D)}. \quad (6.38)$$

Let us now introduce the NP renormalization constant for the tadpole term as follows:  $\Delta_t^2(D) = Z_\Delta \Delta^2$ , where  $\Delta^2$  is finite and positive, by definition, while the NP renormalization constant  $Z_\Delta$  depends on all the unphysical parameters, such as regulating parameters, mentioned above, etc. In principle, the string tension in the linear rising potential between heavy quarks has to be somehow related to the finite  $\Delta^2$ , since it is connected to the transverse projection operator structure, as underlined above. It is well-known that the confining ansatz  $D_{\mu\nu}(q) \sim T_{\mu\nu}(q) \Delta^2 / (q^2)^2$  just leads to a such potential. So that, the ST identity for the renormalized full gluon propagator becomes

$$q_\mu q_\nu D_{\mu\nu}^R(q) = i\xi^R = i \frac{\tilde{\xi}_0 q^2}{q^2 + \tilde{\xi}_0 \Delta^2}, \quad (6.39)$$

where the renormalized gauge-fixing parameters are defined as follows:  $\tilde{\xi}_0 = Z_\Delta \xi_0$ ,  $\xi^R = Z_\Delta \xi$  and  $D_{\mu\nu}^R(q) = Z_\Delta D_{\mu\nu}(q)$ , i.e., the ST identity (6.39) is expressed in terms of the finite quantities only. The correct PT limit is maintained, of course. **The renormalized version of the tadpole term  $\Delta^2$  can be considered as a scale determining the NP structure of the general QCD. Conventionally, we call it as a mass gap** and see discussion in section 10 as well. Such a finite mass gap  $\Delta^2$  will appear in the full gluon propagator as well, i.e., not only in its longitudinal part (see sections 8 and 9).

Let us now demonstrate the interesting feature of the generalized gauge (6.39). It provides the self-consistency condition for the gauge choice in QCD as follows:

$$\frac{\tilde{\xi}_0 q^2}{q^2 + \tilde{\xi}_0 \Delta^2} = \frac{a q^2}{q^2 + a \Delta^2}, \quad (6.40)$$

i.e., the left-hand-side of this equation is present by the generalized gauge expression (6.39), while its right-hand-side presents the same expression when the gauge is already chosen. In other words, we are checking whether the above-mentioned GGD formalism is compatible with its EGD one and *vice versa*. So that its aim is to derive a relation (not an identity) involving the gauge-fixing parameter. If  $a$  is any finite number, then from the self-consistency relation (6.40) it is easy to derive that  $\tilde{\xi}_0 = a$ , indeed. At the same time, if  $a = \infty$  the so-called canonical gauge [14], then the relation (6.40) becomes

$$\frac{\tilde{\xi}_0 q^2}{q^2 + \tilde{\xi}_0 \Delta^2} = \frac{q^2}{\Delta^2}, \quad (6.41)$$

which is only satisfied at  $q^2 = 0$ , i.e., there is no any condition for the gauge-fixing parameter. The formulated self-consistency condition (6.40) points out on the inconsistency of the canonical gauge  $\xi_0 = \infty$  in QCD. Briefly, if everything is expressed in the terms of the finite quantities then the formal canonical gauge is forbidden to use in QCD, according to our approach to this theory.

## 7. Conventional QCD

The widely used up to present days the linear relation  $\xi = \xi_0$  is only a particular solution to the exact constraint (3.15). As mentioned above in section 3, it is defined when the tadpole term has to be put zero by hand, i.e.,  $\Delta_t^2(D) = 0$ , then  $\xi = \xi_0 = \lambda^{-1}$  as well and *vice versa*. The corresponding expressions for the full gluon propagator and the ST identity in this solution to QCD can be independently derived absolutely in the same way as has been derived the corresponding expressions for the general solution to QCD in the previous section. However, the simplest way to show the conventional solution is to put formally  $\Delta_t^2(D) = 0$  in the system of eqs. (5.34)-(5.35), then it looks like

$$D_{\mu\nu}^{PT}(q) = \frac{iT_{\mu\nu}(q)}{q^2[1 + \Pi(q^2; D^{PT})]} + i\lambda^{-1} L_{\mu\nu}(q) \frac{1}{q^2}, \quad (7.42)$$

so that the gluon invariant function is  $d_{PT}(q^2) = [1 + \Pi(q^2; D^{PT})]^{-1}$  and formally neglecting the invariant function  $\Pi(q^2; D^{PT})$ , i.e., putting it zero by hand  $\Pi(q^2; D^{PT}) = 0$ , one obtains the free gluon propagator (3.9). The ST identities in this case are

$$q_\mu q_\nu D_{\mu\nu}^{PT}(q) = q_\mu q_\nu D_{\mu\nu}^0(q) = i\lambda^{-1}. \quad (7.43)$$

The corresponding gluon SD eq. (5.28) now becomes

$$D_{\mu\nu}^{PT}(q) = D_{\mu\nu}^0(q) + D_{\mu\rho}^T(q), \quad (7.44)$$

where

$$D_{\mu\nu}^T(q) = D_{\mu\rho}^0(q) iT_{\rho\sigma}(q) q^2 \Pi(q^2; D^{PT}) D_{\sigma\nu}^{PT}(q). \quad (7.45)$$

From above and now on in this section we denote the corresponding full gluon propagator  $D$  by  $D^{PT}$  (for the explanation see the text below). Obviously, the expression for the PT full gluon propagator (7.42) describes the propagation of the PT massless gluons, since it has the PT singularity on the mass-shell  $q^2 = 0$ , i.e., the singularity of the free gluon propagator  $\sim 1/q^2$ . The gluon SD equation (7.42) does not provide any hint how to prevent the free and the PT massless gluon states to appear at large distance ( $q^2 \rightarrow 0$ ), i.e., this theory is not confining, as emphasized above. Let us note in advance that the equality  $\xi = \xi_0 = \lambda^{-1}$  takes place only for the regularized massless gluon fields. For their renormalized counterparts these gauge-fixing parameters differ by the corresponding renormalization constant.



The conventional or, equivalently, the PT QCD is a well-elaborated quantum gauge field theory [2, 3, 4, 5, 6, 7, 8, 9], and for its framework of the corresponding dynamical equations of motion see recent review [34] (and references therein). Just it has been investigated in full details the last five decades since its formulation in [1]. It is necessary to remind that the coupling constant remains strong, so to call it as  $D^{PT}$  is only a mere convention. This theory is not our concern. We present it here within our formalism for the readers convenience to directly compare it with the general solution to QCD. For this aim it is useful to reproduce the system of the relations (5.36)-(5.37) for the PT QCD. It will look like

$$\begin{aligned} q_\rho q_\sigma \Pi_{\rho\sigma}^q(q) &= q_\rho q_\sigma \Pi_{\rho\sigma}^g(q; D^{PT}) = 0, \\ q_\rho q_\sigma \Pi_{\rho\sigma}(q; D^{PT}) &= 0, \end{aligned} \quad (7.46)$$

and

$$\Delta_g^2 = \Delta_g^2(D^{PT}) = 0, \quad \Delta_t^2(D^{PT}) = 0, \quad \xi = \xi_0 = \lambda^{-1}. \quad (7.47)$$

The first equalities in these relations coincide with the corresponding equalities in the relations (5.36)-(5.37), i.e., in both theories the general and the PT QCD they are exact mathematical results. At the same time, the second equalities in these relations remain prescriptions, since they have to be put zero by hand, as emphasized above.

The dimensional regularization method (DRM) [4, 6, 27, 28, 29], formulated within the framework of the conventional QCD, provides a gauge-invariant scheme to correctly calculate the finite parts of the generally QD loop integrals, while omitting their QD constants, i.e., simply to ignore them and to deal further only with the logarithmic divergences of the PT. This was a prescription rather than an exact result, as pointed out in [4]. However, from now on this prescription has been put on a firm mathematical ground. How to understand such kind of equalities has been already explained in section 5.

## 8. Confinement of the free gluon states

For the sake of transparency and simplicity, we use the free gluon propagator in the t' Hooft-Feynman gauge  $\lambda^{-1} = 1$ , i.e.,  $D_{\mu\nu}^0(q) = i\delta_{\mu\nu}/q^2$ . Then the full gluon propagator (5.28), on account of the definitions (5.29) and (5.30), up to the first iteration term looks like

$$D_{\mu\nu}(q) \sim i \frac{\delta_{\mu\nu}}{q^2} - i \frac{T_{\mu\nu}(q)}{q^2} \Pi(q^2; D) - iT_{\mu\nu}(q) \frac{\Delta_t^2(D)}{(q^2)^2} + \dots, \quad (8.48)$$

where the invariant function  $\Pi(q^2; D)$  is a regular one of its argument, while all the tadpole-like terms up to one loop approximation have been omitted due to the relations (4.26). See appendix A as well, where this iteration is discussed in more detail. Also here and below not

shown the terms proportional to the longitudinal component  $L_{\mu\nu}(q)$  since finally they will be summed up to the known function, see expression (5.34). The higher-order iteration terms will be much more singular with respect to the general ratio  $(\Delta_t^2(D)/q^2)$ , since each iteration invokes additional powers of this ratio. Because of the NL character of the iteration procedure any next iteration will contribute into the previous iteration terms, and thus they become the sums of the infinite number of terms [35], namely

$$D_{\mu\nu}(q) \sim iT_{\mu\nu} \frac{\Delta_t^2(D)}{(q^2)^2} \sum_{k=0}^{\infty} \left( \frac{\Delta_t^2(D)}{q^2} \right)^k \Phi_k(g^2, \dots) + \dots, \quad (8.49)$$

where the terms of order  $O_{\mu\nu}(1/q^2)$  are not shown, for simplicity. In this iteration expansion is explicitly shown only singular part of the corresponding Laurent series, and thus it represents the summation of all the severe IR singularities (more singular than the PT IR singularity  $\sim 1/q^2$ ), which can be accumulated by the full gluon propagator. Evidently, such a summation is only possible due to the existence of the tadpole term. The dependence of the coefficients functions  $\Phi_k(g^2, \dots)$  on the coupling constant is not known, i.e., such kind of series are the cluster expansions, mentioned above [23]. These coefficients, being the sum of the infinite number of terms by themselves, are resulted from the skeleton loop integrals, contributing into the full gluon self-energy through the invariant function  $\Pi(q^2; D)$ . Let us note here that eq. (8.49) can be expressed in terms of the mass gap through the relation  $\Delta_t^2(D) = Z_\Delta \Delta^2$  and further including  $Z_\Delta$  into the above-mentioned coefficient functions. Evidently, the iteration *ab infinity* leads to the essential singularity at  $q^2 \rightarrow 0!$  in the full gluon propagator (8.49), which will remain massless, but depending on the tadpole term (scale breaking in the IR region). Let us note that the same series can be obtained by expanding the initial NL transcendental relation (5.31) in powers of the ratio  $(\Delta_t^2(D)/q^2)$ .

From the NL iteration expressions (8.48)-(8.49) it follows that the full gluon propagator in the IR region ( $q^2 \rightarrow 0$ ) is dominated by the tadpole term contribution, while in the UV region ( $q^2 \rightarrow \infty$ ) it will be dominated by the free gluon one. This is completely different to QED, where the dressed photon propagator behaves at small photon momentum as its free counterpart ( $\sim 1/q^2$ ). This is also true for the full gluon propagator in the conventional QCD, investigated in section 6. The principal difference between these theories is due to the existence of the tadpole term in the QCD ground state, which precisely controls the structure of a single full gluon propagator at small  $q^2$ . That is why the dressed PT gluon (7.42) and free gluon states can not appear in the physical spectrum at large distances ( $q^2 \rightarrow 0$ ), i.e., they will be suppressed in this region by the tadpole term. If this statement will survive the corresponding renormalization program beyond the PT then confinement of the PT and free gluon states will be explained. This is important since there is no any other dynamical mechanism to confine these gluon states to the

vacuum of QCD or to keep them inside hadrons. To prove the confinement of the singular full gluon propagator itself due to the above-mentioned essential singularity, though possible, but requires completely different investigation within the theory of distributions [26]. It is the subject of the future work. Let us note that by the NP program we mean that the full gluon propagator will become free from all the non-physical parameters (like the coefficient functions mentioned above) of the theory in any of its possible solutions (not only in the iteration one).

## 9. Asymptotic freedom

let us now investigate the PT  $q^2 \rightarrow \infty$  limit of the full gluon propagator (5.34). For this aim it is enough to neglect the ratio  $(\Delta_t^2(D)/q^2)$  in the invariant function eq. (5.31). Also, to replace the the invariant function  $\Pi(q^2)$  by its to leading order logarithm, and to introduce the following relation  $d(q^2) = \alpha_s(q^2; \Lambda^2)/\alpha_s(\lambda)$ , where  $\Lambda^2$  and  $\lambda$  are the UV regulating parameter and its dimensionless counterpart, respectively. Then eq. (5.34) will look like

$$D_{\mu\nu}(q) \sim iT_{\mu\nu}(q) \left[ \frac{\alpha_s(\lambda)}{1 + \alpha_s(\lambda)b_0 \ln(q^2/\Lambda^2)} \right] \frac{1}{q^2} + \dots, \quad (9.50)$$

where  $b_0 > 0$  is the colour group factor and  $\alpha_s(\lambda)$  is the coupling constant squared. Here and below the trivial longitudinal part is not shown, for simplicity. Since we already know that the finite mass gap exists, then without loosing generality, one can put  $\Lambda^2 = f(\lambda)\Delta^2 = f(\lambda)a^{-1}a\Delta^2 = f'(\lambda)\Lambda_{QCD}^2$  with  $\Lambda_{QCD}^2 = a\Delta^2$ , where  $a$  is a some finite number because one cannot identify directly the mass gap  $\Delta^2$  with  $\Lambda_{QCD}^2$ . The dimensionless functions  $f'(\lambda) = a^{-1}f(\lambda)$  behave like  $\sim \lambda^2$  in the  $\lambda \rightarrow \infty$  limit, otherwise remaining arbitrary. Doing some algebra, one obtains

$$D_{\mu\nu}(q) \sim iT_{\mu\nu}(q) \left[ \frac{\alpha_s}{1 + \alpha_s b_0 \ln(q^2/\Lambda_{QCD}^2)} \right] \frac{1}{q^2} + \dots, \quad (9.51)$$

where finite  $\alpha_s = \alpha_s(M_Z) = 0.1184$  is determined by the relation connecting  $\alpha_s(\lambda)$  to  $\ln f'(\lambda)$ . Eq. (9.51) has been discovered many years ago [38, 39]. In the PT  $q^2 \rightarrow \infty$  limit the dependence on the coupling constant  $\alpha_s$  vanishes, indeed. The well-known expression (9.51) is written down in the so-called approximation of the main PT logarithms and  $\Lambda_{QCD}^2$  determines the scale of the non-trivial PT dynamics in the QCD.

However, in section 6, we have already shown that the PT QCD cannot generate the scale parameter, having the dimension of a mass squared. From the renormalization group equations analysis, it follows that any mass to which can be assigned any physical meaning is exponentially suppressed in the weak coupling limit to every order of the PT [2, 36], namely

$$M \sim \mu \exp(-1/b_0 g^2), \quad g^2 \rightarrow 0, \quad (9.52)$$

where  $\mu$  is the arbitrary renormalization point. None a finite mass can survive in the PT weak coupling limit or, equivalently, in the PT  $q^2 \rightarrow \infty$  regime, indeed. So the question where the finite mass comes from? cannot be answered by the PT! It is entirely of the NP origin! It has to come from the IR region which is controlled by the tadpole term, as discussed above. So that, the scale breaking appears under the PT logarithm, while the gluon itself remains massless like in eq. (8.49), where the scale breaking appears in the NL way. In the both cases the scale breaking masses cannot be directly identified with the mass gap, defined in section 6. It should be multiplied by the finite numbers different in each case, in order to get scale breaking in the IR and UV regions. Its exact derivation, and thus the explanation of the AF phenomenon (i.e., a scale breaking in the PT regime) will be given when the corresponding NP renormalization program for the full gluon propagator (5.34) will be performed.

Concluding, all this can be manifestation that "the problems encountered in perturbation theory are not mere mathematical artefacts but rather signify deep properties of the full theory" [43]. The message we are trying to convey is that the AF phenomenon is the first direct evidence of the validity of the general QCD. In our opinion, the gluon confinement and AF are the two boundary conditions at low- and high-energies in the QCD and its vacuum, to which the general QCD satisfies, while the conventional QCD does not. For these features of the general QCD, one can call it as the confining QCD theory as well. Finally, it is worth emphasizing once more that in the both asymptotics of the iteration/singular solution to the confining QCD, the gluons always remain massless. So that, the QCD Lagrangian's gauge symmetry is not violated by it, but the mass gap is needed in order to provide its scaling properties, as described above.

## 10. Discussion and outlook

The general QCD requires the explicit presence of the mass squared scale parameter - the tadpole term - in the QCD ground state, i.e., at the fundamental quark-gluon level, as this was exactly proven in section 4 and used in section 5. In its presence the role of the QCD coupling constant  $g^2$  becomes unimportant. This is also evidence of the 'dimensional transmutation',  $g^2 \rightarrow \Delta_t^2(D)$  [2, 36, 37], which occurs whenever a massless theory acquires mass dynamically. It is a general feature of spontaneously symmetry breaking in field theories. We distinguish between the PT/particular/conventional and general QCD by the explicit presence of the tadpole term in the latter one, and not by the magnitude of the coupling constant even at the regularized gluon fields level yet. In the both cases the gluon fields remain strongly interacted, apart from the AF regime [38, 39].

The general character of our results is convenient to summarize in connection with the Jaffe-Witten's (JW) [23] theorem, formulated as follows:

**JW's theorem: Yang-Mills existence and Mass Gap.** Prove that for any compact simple gauge group  $G$ , a non-trivial quantum Yang-Mills theory exists on  $R^4$  and has a mass gap  $\Delta > 0$ . Existence includes establishing some strong axiomatic properties for the Euclidean Green's functions.

In their description of the theorem it is explained why a correct QCD theory must have the following three properties: 1). The nuclear force is strong but short-ranged due to the existence of the "mass gap", a scale parameter  $\Delta > 0$ . 2). The physical particle states are  $SU(3)$ -invariant, i.e., confinement of colour degrees of freedom at the fundamental quark-gluon level. 3). The chiral symmetry breaking to account for the "current algebra" theory of soft pions at the hadronic level.

The definition of the mass gap given in [23] as the solution of the Hamiltonian problem that every excitation of the vacuum has energy at least  $\Delta > 0$  is the mass gap responsible for the second PCAC phase transition at the hadronic level (in our opinion). For example, the glueball state with the most lowest mass, though none of such state has been discovered yet (but this is not a problem, reminding that the Higgs particle [41] has been experimentally found after many years of its prediction). The literary meaning/definition of the mass gap, mentioned above, may appear at the fundamental quark-gluon level as well in the general QCD only in its massive solution for the full gluon propagator (5.34). The gluon may acquire mass dynamically, indeed, if the denominator of this equation becomes zero at some finite point  $q^2 = M^2 = \Delta_t^2(D)$ . This solution is completely different case from the singular solution, and will be subject for the independent consideration.

The request for a mass gap in the general QCD, can be also summarized like the JW theorem:

**Theorem I: Mass Gap existence and Yang-Mills theory.** If a non-trivial quantum Yang-Mills theory with gauge group  $SU(3)$  exists on  $R^4$  then it has a mass gap  $\Delta^2 > 0$ , responsible for the first confinement phase transition in QCD. Existence includes establishing the confining QCD theory, which remaining the PT renormalizable.

In connection with our theorem, let us note that the tadpole term cannot be removed from the QCD theory and its vacuum by any means, apart from putting it zero by hand in order to get to the conventional QCD. Its renormalized (finite) version survives the NP renormalization and thus can be assigned a physical meaning as a scale parameter responsible for the NP effects in the QCD, leading to the formulation of the confining QCD theory. Just the existence of a such scale, conventionally called as a mass gap, provides the suppression of the PT and free gluon states to appear at large distances, i.e., in the physical spectrum, as explained above. It is dynamically generated by the self-interaction of the multiplied massless

gluon modes, dominated by the point-like four-gluon vertex [10, 23, 40], i.e., it is a fundamental quantity ("mass without mass" [44]), so that it cannot be a bound state of anything. In [10] the importance of the mass gap has been recognized and exploited on an intuitive basis rather than a logical one, but from now on its existence cannot be undermined.

A few additional clarifying remarks are in order. Let us remind that the present QCD theory, i.e., conventional one in our terminology, possesses none of the above-mentioned three necessary properties required by JW for QCD to be successful theory of the strong interactions at the both microscopic and macroscopic levels. Unlike this theory, the confining QCD theory, discovered in this work, possesses at least the two of those properties. It has a characteristic mass scale parameter even at the fundamental quark-gluon level. Precisely, the existence of this scale in the theory confines the colour gluons to the QCD ground state. We strongly believe that quark confinement and "chiral symmetry breaking" will be also confirmed after solution of the quark full SD equation by the substitution of the full gluon propagator, determined within the confining QCD theory. It is worth emphasize once more that we have explicitly shown that the first confinement phase transition in QCD should have its own finite scale parameter  $\Delta^2$ , reflected in our **Theorem I**. In order to describe the scales of the different NP effects, such as the scales breaking, the existence of the string tension, etc., it should undergo the finite transformations, i.e., to be multiplied by the different finite constants.

Taking into account that the JW theorem has been formulated in the most general terms, absorbing the different interpretations/meanings/understanding of a mass gap itself, it was quite natural and suitable for us to adopt the JW mass gap name for our characteristic mass scale parameter. In other words, we understand the mass gap conception in more wider context than its definition given by JW. Nevertheless, to summarize all our advance results in terms of the theorem similar to the JW theorem, makes sense and is very useful for further development of the correct (confining) QCD theory and understanding of its true vacuum dynamical and gauge structures. In obtaining them no any truncations, approximations and assumptions have been made in dealing with the regularized skeleton loop integrals, contributing into the full gluon self-energy. The exact analytical derivations have been done, based on a tensor algebra rigorous rules only.

The perspectives for further work are as follows: the NP renormalization program of the ST identity performed in section 6 was enough to prove the existence of a mass gap, as stated in **Theorem I**. However, to formulate the NP multiplicative (MP) renormalization program for the regularized full gluon propagator (5.34) is not a simple task. Such kind of a program has to take into account of all the peculiarities of the severely singular gluon field configurations, briefly discussed in section 8: first of all, the transcendental character and the NL structure of eq. (5.31).

It has to be replaced by the corresponding cluster expansion, suitable for the NP renormalization programs to be performed. This is also needed to be formulated in the most general and unique ways. To ensure a unitarity of  $S$ -matrix elements by these NP MP renormalization program should be requested as well. In general, our results open a new wide research areas in the general QCD, leading to the multiple theoretical and phenomenological applications and comparing them with the results of the conventional QCD. In this connection, let us note that the general QCD may have another type of the solution, different from the singular one, due to the above-underlined transcendental character and the NL structure of eq. (5.31), namely the massive gluon solution, mentioned above.

To understand a mass dynamical generation at the quark-gluon level is the first necessary step to understand the existence of the physical mass spectrum at the hadronic level in QCD as a theory of strong interactions. The confinement first and only after the PCAC phase transitions in QCD have to be understood [2, 23]. QCD is a self-consistent quantum gauge field theory. It does not need any extra degrees of freedom (such as Higgs fields [41]) or to extract the mass by some other ways to generate a mass at the fundamental quark-gluon level.

The relevant existing knowledge on a mass dynamical generation and spontaneously symmetry breaking subjects has been taken into account in the list of publications, presented in our paper (and see references in each of the cited items there). Let us note that contrary to the ghost term, the QCD Lagrangian's gauge-fixing term is not explicitly present in the full gluon self-energy, that is why it is left out of our consideration.

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## Appendix A. The full gluon propagator up to one loop

In order to explicitly compare the tadpole term (2.4) with the other tadpole-type terms let us consider the iteration of the gluon SD eq. (2.1), shown in Fig. 1, up to one loop, shown in Fig. A.2. The last two terms in this figure appear because of the NL nature of the iteration series, when the next iterations contribute into the previous ones. That is the reason why such multiplied tadpole-type terms are not present in the skeleton loop decomposition for the

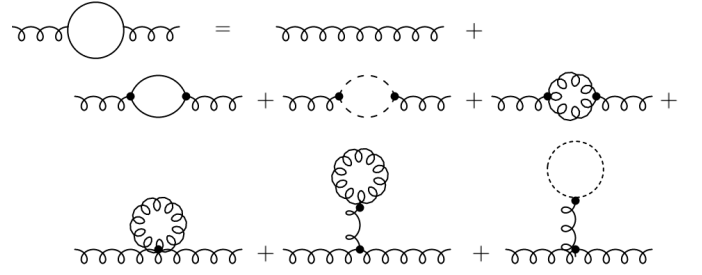


Figure A.2: The SD equation for the full gluon propagator up to one loop as present in [6]. It is useful to compare it with Fig. 1 and its description in the text after it.

full gluon propagator in Fig. 1, but have to be shown up in the iteration terms, seen in Fig. A.2. This iteration has been exactly calculated in [6] by using the DRM, while rightfully now ignoring the tadpole term itself and all the other tadpole-like ones, in accordance with the relations (7.47). It is worth emphasizing that in [6] this iteration has not been called the PT expansion up to  $O(g^2)$ , though looks like it, but expansion up to  $O(\hbar)$ . It is well-known that in QCD the expansion in powers of the coupling constant does not make any sense because it is strong. The PT expansion in QCD makes sense only in the AF [37, 38] regime when it becomes weak. In order to perform analytical derivations below, we will take the interaction vertices as the point-like ones. The external gluon propagators will be taken as the free ones. All the skeleton loop propagators will remain as the full ones. Then the corresponding iteration will look like Fig. A.2, and hence can be called as quasi-one loop expansion, discussed in some details in section 8, when it was called as the cluster one. Note that below we do not show explicitly the dependence on the coupling constant, since our skeleton loop quark, gluon and ghost propagators are the full ones, so that the exact dependence on the coupling constant of the expressions below is not known. Also we will omit below all the colour group factors and other overall numerical numbers, since all of them are unimportant here.

However, it is instructive first to show explicitly how works our subtraction scheme, using only the third diagram in the second line of Fig. A.2, for simplicity. From [6] by replacement  $\mu \rightarrow \rho$  and  $\nu \rightarrow \sigma$  and fixing  $\omega = 2$  and  $\delta_{\rho\rho} = 4$  in eq. (6.22) there, one finally obtains an analytical expression for this term as follows:

$$\Pi_{\rho\sigma}^{(3)}(q) \sim \int d^4l \frac{N_{\rho\sigma}(l, q)}{l^2(l+q)^2}, \quad (\text{A.1})$$

where, apart from below, we put the gluon loop propagators as free ones in the  $t'$  Hooft-Feynman gauge as well, for the sake of transparency of our derivations for this diagram. The tensor function  $N_{\rho\sigma}(l, q)$  is

$$N_{\rho\sigma}(l, q) = 10l_\rho l_\sigma + 5(l_\rho q_\sigma + q_\rho l_\sigma) - 2q_\rho q_\sigma$$

$$+ (5q^2 + 2(lq) + 2l^2)\delta_{\rho\sigma}. \quad (\text{A.2})$$

Putting in eqs. (A.1) and (A.2)  $q = 0$ , one gets

$$\Pi_{\rho\sigma}^{(3)}(0) \sim \int d^4l \frac{N_{\rho\sigma}(l, 0)}{(l^2)^2} = \int d^4l \frac{(10l_\rho l_\sigma + 2l^2 \delta_{\rho\sigma})}{(l^2)^2}, \quad (\text{A.3})$$

and, obviously, this integral is quadratically divergent at the upper limit, but is assumed to be regularized from above and below from the very beginning, as emphasized throughout this work.

Let us now compose the subtracted counterpart of the initial integral (A.1), in accordance with the subtraction (4.18), as follows:

$$\Pi_{\rho\sigma}^{(3s)}(q) = \Pi_{\rho\sigma}^{(3)}(q) - \Pi_{\rho\sigma}^{(3)}(0) \sim \int d^4l \frac{N_{\rho\sigma}^s(l, q)}{(l^2)^2(l+q)^2}, \quad (\text{A.4})$$

where

$$\begin{aligned} N_{\rho\sigma}^s(l, q) &= 5l^2(l_\rho q_\sigma + q_\rho l_\sigma) - 20(ql)l_\rho l_\sigma - 2l^2 q_\rho q_\sigma \\ &+ l^2(q^2 - 2(lq))\delta_{\rho\sigma} - 10q^2 l_\rho l_\sigma. \end{aligned} \quad (\text{A.5})$$

Because of the last term and terms  $l^2 q^2 \delta_{\rho\sigma}$ ,  $2l^2 q_\rho q_\sigma$  in this equation, the subtracted integral (A.4) is only logarithmically divergent at the upper limit. So that the integral (A.3) is one of the tadpole-like terms which should be disregarded on the general basis due to the relations (4.26) within our approach. The subtracted integral (A.4) finally will contribute to the invariant function which appear in the full gluon propagator (5.34).

The tadpole term up to one skeleton loop, shown first in the third line of Fig. A.2, analytically can be written down as follows:

$$\Pi_{\rho\sigma}^t(D) \sim \int d^4l T_{\rho\sigma\alpha\beta}^0 D_{\alpha\beta}(l) = \delta_{\rho\sigma} \Delta_t^2(D), \quad (\text{A.6})$$

see eq. (2.4) as well. Being the QD but already regularized constant, it is not connected to any external gluon momentum. In this respect it is different from all other tadpole-like terms which appear as a result of the subtraction scheme or in the NL iteration procedure. We have shown that in the conventional QCD (described in section 6) it was a prescription to neglect it, but in the confining QCD (investigated in sections 4 and 5) it should remain intact. At the same time, all the other tadpole-like terms, mentioned above, have to be disregarded on the exact mathematical basis in the both theories.

Let us now investigate the tadpole-like term with the gluon skeleton loop, shown second in the third line of Fig. A.2. Its analytical expression is

$$\Pi_{\rho\sigma}^{(4)}(q, p) \sim T_{\rho\sigma\mu'}^0(q, p) D_{\mu\mu'}^0(p) \int d^4l T_{\nu\zeta\mu}^0(l, p) D_{\nu\zeta}(l), \quad (\text{A.7})$$

where the gluon momentum  $p$  is, in fact, zero, i.e.,  $p = 0$  but it is convenient to go to this limit at the final step only. Both gluon propagators in the t' Hooft-Feynman gauge are  $D_{\mu\mu'}^0(p) = \delta_{\mu\mu'}/p^2$  and  $D_{\nu\zeta}(l) = \delta_{\nu\zeta} d(l^2)/l^2$ , where  $d(l^2)$

is the corresponding invariant function of the full gluon propagator. Let us remind that such kind of the skeleton loop integrals are assumed to be regularized from above and below. Using now the Euclidean space Feynman rules for the corresponding vertices present in [6], one arrives at

$$T_{\rho\sigma\mu'}^0(q, p) = -(2q+p)_{\mu'}\delta_{\rho\sigma} + (q-p)_\sigma\delta_{\rho\mu'} + (2p+q)_\rho\delta_{\sigma\mu'} \quad (\text{A.8})$$

and

$$T_{\nu\zeta\mu}^0(l, p) = 2(l+p)_\mu\delta_{\nu\zeta} - l_\zeta\delta_{\nu\mu} - (2p+l)_\nu\delta_{\zeta\mu}. \quad (\text{A.9})$$

Substituting these expressions into the eq. (A.7), one finally obtains

$$\Pi_{\rho\sigma}^{(4)}(q, p) \sim T_{\rho\sigma\mu}^0(q, p) \frac{2}{p^2} \int d^4l \frac{d(l^2)}{l^2} [3l_\mu + p_\mu]. \quad (\text{A.10})$$

By the symmetry integration ( $d^4l = l^3 dl = (1/2)l^2 dl^2$  and all the overall numerical numbers due to the integration over angular variables in four-dimensional Euclidean space [6] omitted) the first loop integral is zero, while the second one leads to

$$\Pi_{\rho\sigma}^{(4)}(q, p) \sim T_{\rho\sigma\mu}^0(q, p) p_\mu \frac{2}{p^2} \times \tilde{\Delta}_g^2(d), \quad (\text{A.11})$$

where  $\tilde{\Delta}_g^2(d) = \int d^4l (d(l^2)/l^2)$  is the QD but regularized skeleton loop integral, having the dimension of a mass squared. It is one of the constants present in the relations (4.26). It should be discarded on the general mathematical basis within our approach, i.e., put zero, so that  $\tilde{\Delta}_g^2(d) = 0$ , even before the final  $p^2 = -M^2 \rightarrow 0$  limit. Let us note that the cancellation of the pole  $1/p^2 = -1/M^2$  in eq. (A.11) can be always achieved by going to the dimensionless variable  $x = l/M$  in the skeleton loop integral (A.10), i.e., this pole is not a problem here. Thus,  $\Pi_{\rho\sigma}^{(4)}(q, p) = 0$ .

Let us finally investigate the tadpole-like term with the ghost skeleton loop, shown third in the third line of Fig. A.2. In analogy with eq. (A.7) its analytical expression is

$$\Pi_{\rho\sigma}^{(5)}(q, p) \sim T_{\rho\sigma\mu'}^0(q, p) D_{\mu\mu'}^0(p) \int d^4l \frac{l_\mu}{l^2} d_{gh}(l^2), \quad (\text{A.12})$$

where  $d_{gh}(l^2)$  is the invariant function for the full ghost propagator  $G(l)$ . The vertex  $T_{\rho\sigma\mu'}^0(q, p)$  is explicitly shown in the relations (A.8), and the remaining free gluon propagator is also the same. The altered rule [6] for the gluon-ghost vertex gives  $(1/2)[(l_\mu + l_\mu) = l_\mu$  and the gluon momentum  $p$  is zero again at the final stage. Since the nominator of this skeleton loop integral contains the loop variable linearly, then by the symmetry integration it is simply zero from the very beginning, i.e.,  $\Pi_{\rho\sigma}^{(5)}(q, p) = 0$ .

Concluding, neither the tadpole-like term with the gluon skeleton loop nor its ghost counterpart contribute into the full gluon propagator. In these cases the subtractions with respect to the external gluon momentum  $q$  (though possible), but were not even necessary to perform, since the

final results would have been the same zeros. The primary goal of this work is to prove that the tadpole term cannot be disregarded by any means. All the other tadpole-like terms are to be always discarded on the general basis, i.e., put zeros, as explicitly demonstrated here as well. This was not a prescription, but being an exact mathematical result within our formalism.

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